Cloud Strategy Proposal for Northbridge Re: Executive Summary and Assessment Report

Executive Summary

For this report I chose a hypothetical reinsurance company, which I have called Northbridge Re. It is based in London, writes property catastrophe and life reinsurance for insurers in the UK and Europe, and has around 800 staff. Like a lot of firms in this sector, it still runs most of its core systems from two on premises data centres, including its treaty administration platform, actuarial modelling tools and a data warehouse used for regulatory reporting.

The company has three main problems. The first is catastrophe modelling. After a major event, or during the January renewal season, it needs a large amount of computing power for a few weeks, but it must own enough hardware to cover that peak all year round. The second is that new environments take weeks to build because servers are set up by hand, so environments are rarely identical. The third is that underwriters and actuaries want to use machine learning on claims data, but the data is spread across several systems and is hard to access.

My proposed solution is a hybrid cloud strategy using Amazon Web Services (AWS). The infrastructure is written as code, with Terraform creating the AWS resources and Ansible configuring the servers. Workloads that need to scale, such as catastrophe modelling and analytics, move to AWS first. Older systems that would be expensive to move, and data the company prefers to keep in house, stay in the existing data centre and connect to AWS over a private link.

I expect three main outcomes. Model runs should take hours rather than days, costs should shift from fixed hardware spend to paying for what is used, and every change to the infrastructure should be recorded, which helps with audits in a regulated sector. These match the main drivers of cloud adoption that Alshareef (2023) identifies, namely scalability, lower costs and faster deployment.

Cloud Solution Design

Architecture overview

Figure 1 shows the overall architecture, including the link to the on-premises data centre and the pipeline used to deploy changes.

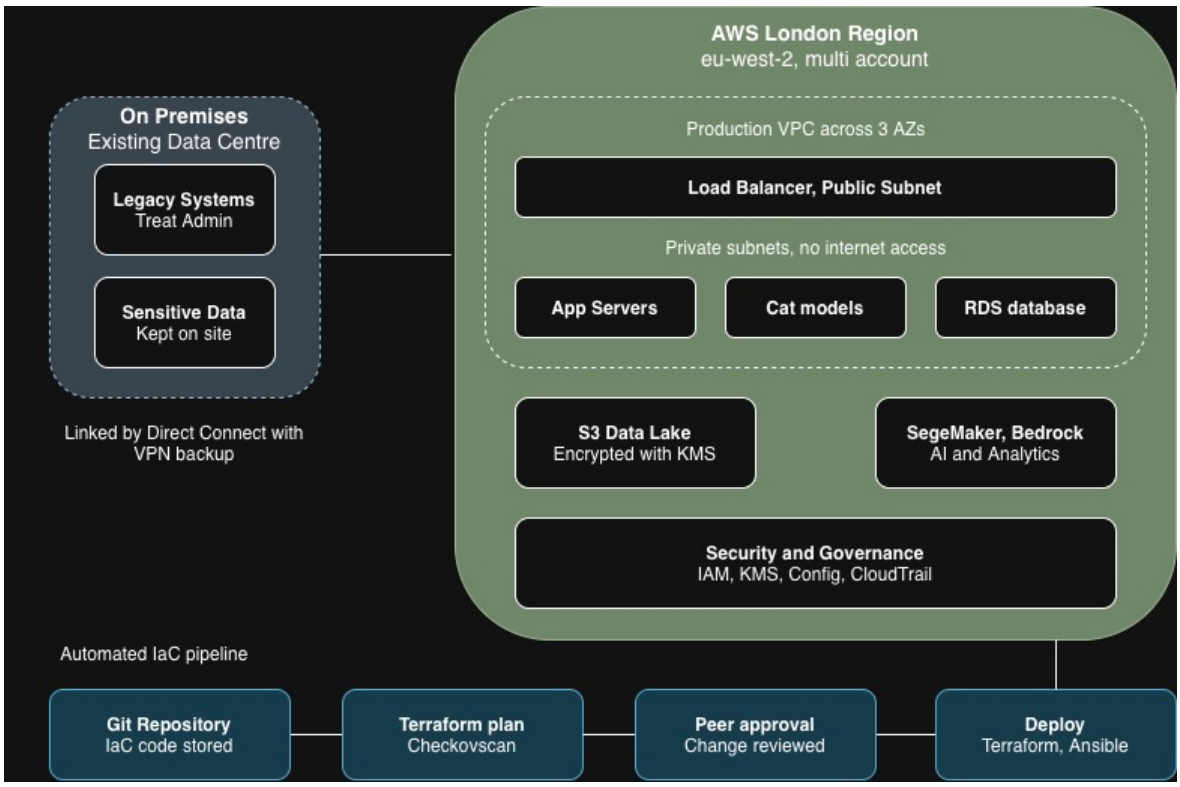


Figure 1: Proposed hybrid cloud architecture for Northbridge Re (Draw.io diagram).

I used separate AWS accounts for production, non-production, security and shared services, managed through AWS Organizations. I did this so that a problem in one account is less likely to affect the others, and so that costs are easier to track. Everything runs in the London region to keep data in the UK.

In the production account, the network (a VPC) is spread across three availability zones. Only the load balancer sits in a public subnet. The application servers, the catastrophe modelling cluster and the databases sit in private subnets with no direct internet access. Data is stored in Amazon S3 and Amazon RDS and is encrypted using keys held in AWS Key Management Service (KMS). The on-premises data centre connects through AWS Direct Connect, with a VPN as a backup.

Using Terraform and Ansible together

I chose both tools because they do different jobs. Terraform is good at creating and changing cloud resources such as networks, storage and servers, and it keeps a record of what it has built (HashiCorp, 2024). Ansible is better at what happens once a server exists, such as installing software and applying security settings (Red Hat, 2024). Morris (2020) explains that the main benefit of Infrastructure as Code is that infrastructure becomes repeatable and

can be reviewed and version controlled like normal code, and that was the main reason I wanted to use it here.

The Terraform example below creates the data lake bucket and blocks public access. I based it on the AWS provider documentation (HashiCorp, 2024).

```
# Data lake bucket with public access blocked
# Based on HashiCorp (2024) AWS provider documentation
resource "aws_s3_bucket" "data_lake" {
  bucket = "northbridge-re-data-lake"
  tags   = { CostCentre = "analytics" }
}

resource "aws_s3_bucket_public_access_block" "data_lake" {
  bucket                  = aws_s3_bucket.data_lake.id
  block_public_acls       = true
  block_public_policy     = true
  ignore_public_acls      = true
  restrict_public_buckets = true
}
```

Once the servers exist, Ansible applies the same basic hardening to each one. This is a shortened version.

```
# Basic hardening for modelling servers
# Based on Red Hat (2024) Ansible documentation
- name: Harden modelling servers
  hosts: cat_modelling
  become: true
  tasks:
    - name: Disable root login over SSH
      ansible.builtin.lineinfile:
        path: /etc/ssh/sshd_config
        regexp: "^PermitRootLogin"
        line: "PermitRootLogin no"
```

Scalability, security and cost

For scalability, the modelling servers sit in an autoscaling group, so more servers are added during busy periods and removed afterwards. Because the environment is defined in code, I could also rebuild it in another region if needed.

For security, I followed the AWS Well Architected Framework, which focuses on least privilege access, encryption and logging (Amazon Web Services, 2024). Putting security settings in the Terraform code means a bucket cannot be created without them. Rahman, Parnin and Williams (2019) found that IaC scripts often contain security problems such as hard coded passwords, so I added a scanning step before any code is applied, which I explain in Section 3.

For cost, every resource has a cost centre tag so spending can be traced. Catastrophe model runs can be stopped and restarted, so they can use Spot Instances, which are much cheaper. Systems that run all the time are covered by Savings Plans, and test environments are switched off outside working hours.

Integration of Advanced Cloud Technologies

Artificial intelligence

I think there are two realistic early uses of AI for Northbridge Re. The first is reading documents. Treaties and claims submissions are long and unstructured, and staff spend a lot of time picking out terms such as limits and exclusions. A generative AI service through Amazon Bedrock could pull out these terms for an underwriter to check. The second is claims analysis, where machine learning models built in Amazon SageMaker could help spot unusual claims or improve reserving estimates.

Carugati (2023) points out that the large cloud providers are also major suppliers of generative AI, and that this link between the two markets could strengthen the position of a few big companies. For Northbridge Re, this made me think about lock in. If its AI tools depend too heavily on one provider, moving away later would be difficult and expensive. To reduce this, I would store data in open formats and use Bedrock because it gives access to models from several companies. I would also keep a person reviewing AI outputs, since Lu et al. (2024) stress the importance of human oversight and clear accountability, and here AI could affect pricing and claims decisions.

Automation

All the Terraform and Ansible code is stored in Git. When someone makes a change, a pipeline runs `terraform plan`, scans the code with Checkov and waits for a second person to approve it before anything is applied. This gives a clear record of who changed what. AWS Config then checks the live environment and flags anything changed by hand. This links back to the Twelve Factor principles I looked at earlier in the module, where keeping configuration separate from code and keeping environments alike both become easier when deployment is automated (Wiggins, 2017).

Hybrid cloud

I chose hybrid rather than moving everything at once. Some legacy systems would need a lot of rework to run in the cloud, and the business may want to keep some sensitive data on site for now. Sharma and Kumar (2024) describe hybrid cloud to balance flexibility with control by placing each workload where it fits best. In my design, the on-premises systems send data to AWS through Direct Connect, and the same Ansible playbooks are used on both sides so servers are hardened to the same standard wherever they run.

Risk and Compliance Considerations

As a UK reinsurer, Northbridge Re is regulated by the Prudential Regulation Authority (PRA) and the Financial Conduct Authority (FCA). The PRA treats significant cloud use as outsourcing and expects firms to manage concentration risk and have a tested exit plan (Prudential Regulation Authority, 2021). The FCA has similar expectations around oversight and security (Financial Conduct Authority, 2019). Because the company has European clients, the Digital Operational Resilience Act (DORA) is also relevant (European Union, 2022), and health data in life reinsurance falls under UK GDPR (Information Commissioner's Office, 2023).

The table below shows the main risks I identified.

Risk	Impact	Mitigation
Misconfigured resources exposing data	High	Security settings in Terraform, Checkov scanning, AWS Config
Vendor lock in	High	Exit plan, open data formats, reusable Ansible playbooks
Weak access control	High	Least privilege roles, multi factor authentication
Incorrect or biased AI outputs	Medium	Human review, monitoring of models
Personal and health data mishandled	High	Encryption, UK region, data protection impact assessments
Cost overruns	Medium	Tagging, budgets and alerts
Regional outage	Medium	Three availability zones, rebuild from code

Table 1: Main risks identified

For me, lock in is the most important risk because it connects the technical design to the regulator's expectations. The PRA expects firms to show they could leave a provider, not just say they could. The Ansible playbooks would move with the company, but the Terraform code is written for AWS and would need rewriting for another provider, so I do not want to overstate how portable this design is.

Another risk is the shared responsibility model. AWS secures the underlying infrastructure, but Northbridge Re is still responsible for its own settings, user access and data (Amazon Web Services, 2024). Many cloud breaches come from customer mistakes rather than provider failures, which is why my design tries to enforce the correct settings automatically.

Future Recommendations

I would recommend that Northbridge Re watches three areas.

The first is edge computing. Shi et al. (2016) describe it as processing data closer to where it is created, which reduces delay and the amount of data sent to central systems. As insurers use more sensor data, such as flood sensors, processing some of it at the edge could make it possible to monitor exposure in near real time after an event.

The second is quantum computing. Amazon Braket already lets companies try quantum hardware through the cloud (Amazon Web Services, 2025). Catastrophe modelling might benefit one day, but I think a small, low cost trial is the most that makes sense for now. The more urgent issue is security, because future quantum computers may break some of today's encryption. NIST has published its first post quantum encryption standards (National Institute of Standards and Technology, 2024), and since reinsurance data is kept for many years, I would suggest the company starts identifying where it uses encryption that could be at risk.

The third is cost and sustainability. Alshareef (2023) lists energy use and cost control as ongoing challenges in cloud computing. Reporting the carbon footprint of workloads and sharing cost data with business teams would help the company manage both.

Conclusion

In this report I proposed a hybrid cloud strategy for Northbridge Re that moves its most demanding workloads to AWS while keeping a secure link to the systems that stay on site. Using Terraform and Ansible means the infrastructure can be rebuilt, checked and audited, which gives a good base for bringing in AI carefully. From working through this, I found that the hardest parts were not technical but around regulation, especially exit planning and avoiding too much dependence on one provider as cloud and AI become more closely linked (Carugati, 2023). Starting with catastrophe modelling would let the company show the value of the cloud early while keeping the risk manageable.

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